Some first-order theory for the random effects meta-analysis

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Abstract. The inverse-variance weighted approach to random effects meta-analysis comprises a number of methods, differing primarily in their handling of the variance component τ^2. Several variant estimators of τ^2 remain in popular use. An ostensibly different approach to random effects meta-analysis uses the likelihood ratio test for inference. We verify that, in the vicinity of the null hypothesis of no treatment effect, the likelihood ratio test is, up to second order error, the same as the random effects inverse-variance test when the frequently recommended REML estimator of τ^2 is used. Working within a double asymptotic framework, we show more generally that differences between the two tests previously observed are attributable primarily to differences in the choice of τ^2 estimator. After establishing the similarity between the inverse-variance weighted and likelihood ratio tests, we study the underlying common test. We examine the consistency and rates of the tests under the null hypothesis, giving relatively general conditions under which the tests achieve $O(1/n)$ convergence rates.

Keywords: Meta-analysis, Delta method, Edgeworth expansion.

1 Introduction

In the world of the random effects meta-analysis model, scientists studying a common intervention are all unwittingly working with pertrubed data. Even supposing the intervention has no effect, their p-values deviate somewhat from uniformity, with the magntiude of the deviation identified with a parameter τ^2. Sometimes confidence intervals are centered too high and sometimes too low, though on balance they are centered correctly. It takes a meta-analysis to discern this central truth. The meta-analyst works with n pairs of data $(Y_1, \hat{\sigma}_1), \dots, (Y_n, \hat{\sigma}_n)$, with Y_i respresenting the effect estimate produced by study i and $\hat{\sigma}_i$ its standard error. The most common procedure is inverse-variance weighting ("IV"). The meta-analyst first obtains an estimate $\hat{\tau}^2$ of τ^2 by any of several means, and next forms the statistic

$$T_{IV} = \frac{\sum_{i=1}^n w_i(Y_i - \mu_0)}{\sqrt{\sum_{i=1}^n w_i}}$$
$$w_i = 1/(\hat{\tau}^2 + \hat{\sigma}_i^2).$$

This statistic serves as a z-statistic for inference on μ, the underlying effect measured by the Y_i, under the null hypothesis $\mu = \mu_0$.

Two prominent issues with the IV test are the following.

1. The number n of independent samples that go into a meta-analysis is typically small (Davey et al., 2011) for estimating a variance component. This challenge has prompted the introduction of numerous estimators of τ^2. While the DerSimonian-Laird (DerSimonian and Laird, 1986) and restricted maximum likelihood estimators appear to be most popular, many of the others are thought to be superior in specific settings (Sidik and Jonkman, 2007).

2. The IV testing procedure, in carrying out inference on μ_0, ignores the estimation error of $\hat{\tau}^2$. Hardy and Thompson (1996) proposed using a likelihood ratio test to estimte μ and τ^2 simultaneously. This LR test is based on the heteroscedastic Gaussian log-likelihood

$$l(\mu, \tau^2) = -\frac{1}{2}\sum_{i=1}^{n} \log(2\pi(\hat{\sigma}_i^2 + \tau^2)) - \frac{1}{2}\sum_{i=1}^{n} \frac{(Y_i - \mu)^2}{\hat{\sigma}_i^2 + \tau^2}. \tag{1}$$

   The test statistic for the null $\mu = \mu_0$ is

$$T_{LR} = T_{LR}(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}, \hat{\tau}^2_{MLE,R}) = 2(l(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}) - l(\mu_0, \hat{\tau}^2_{MLE,R})) \tag{2}$$

   where $(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE})$ maximizes $l(\mu, \tau^2)$ and $\hat{\tau}^2_{MLE,R}$ maximizes $\tau^2 \mapsto l(\mu_0, \tau^2)$. The test statistic is referred to a chi-squared distribution, with 1 degree of freedom.

We find that the solution proposed for the second problem, the likelihood ratio test, may be viewed as a solution to the first problem, finding a good estimator of τ^2 for the IV test. Specifically, letting $T_{IV}(\hat{\tau}^2_{REML})$ refer to the inverse-variance weighted test statistic using the REML estimate of τ^2, we establish in Section 2.1 that $T_{IV}(\hat{\tau}^2_{REML})^2 = T_{LR} + O(1/n)$ when the null $\mu = \mu_0$ is at least approximately true. Since a two-sided z-test with test statistic Z is the same as a chi-squared test on Z^2, p-values and CIs obtained using the IV test and LR test are the same up to second order. More generally we bound the gap between $T_{IV}(\hat{\tau}^2)$ and T_{LR}, for a generic estimator $\hat{\tau}^2$, in terms of the gap between $\hat{\tau}^2$ and the MLEs $(\hat{\tau}^2_{MLE}, \hat{\tau}^2_{MLE,R})$.

The preceding results don't depend on the usual random effects statistical model or much statistical modeling at all, being algebraic in nature. To continue the analysis, we adopt a statistical model in Section 2.2. This model is somewhat more general than many previous models of random effects meta-analysis. The study variances are viewed as random and subject to finite-sample error, and we introduce a parameter $m = m(n)$ tracking the asymptotic growth of the primary study sizes alongside n. This more general framework lets us describe the relationship between $T_{IV}(\hat{\tau}^2)^2$ and T_{LR} as $O_P(1/m) + O_P(1/n)$ for typical τ^2 estimators. In Section 2.3 we then examine the consistency and rates of the tests. We find that $n = o(m^2)$ is a key condition for consistency, while $n = O(m)$ leads to $O(1/n)$ convergence of confidence intervals. These theoretical findings are then examined by simulation in Section 3.

2 Theory

In Section 2.1 we study the differences between the likelihood ratio and inverse-variance weighted tests, relating it to the choice of τ^2 estimator and the distance between the data and the null hypothesis. While the results are non-parametric in nature, questions of modeling arise naturally, in particular the relationship between the number of studies and the sizes of those studies. In Section 2.2 we therefore introduce a double asymptotic model for random effects meta-analysis. We attempt to formulate a more realistic model than the "working models" usually employed. In Section 2.3 we study the basic properties of the IV and LR tests under this statistical model.

Notation: We use P_n for the empirical mean, i.e., sample average. We use E_n, Var_n, and Cov_n to refer to average moments, e.g., $E_n(YV) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}(Y_i V_i)$.

2.1 Similarity between the IV and LR tests

There is striking visual agreement between the LR test and a two-sided test based on the IV statistic. Figure 1 uses synthetic meta-analysis data to track the convergence of the p-values of each of the two tests to their nominal values. This is the comparison of interest to a meta-analyst who is choosing between the two tests and is concerned about a small sample size or efficiency in general. The figure suggests that, as the number of studies n grows, each statistic converges to its reference distribution, chi-squared in the case of the LR test and standard normal for the IV test, at about the same rate. At each n the depicted simulation uses n^3 monte carlo repetitions, yet even the resulting $O(n^{-3/2})$ monte carlo error is comparable to the gap between the two tests. The similarity betwen the LR and IV tests appears to hold near the boundary $\tau^2 = 0$ of the random effects model and also at alternatives near the null $\mu = \mu_0$. It appears to hold whether or not the data looks Gaussian and in fact the data in Fig. 1 have been chosen to be quite irregular; see Section 3 for simulation details.

The algebraic identity between the IV and LR statistics emerges once we plug the same estimator for $\hat{\tau}^2_{MLE}$ and $\hat{\tau}^2_{MLE,R}$, the full and constrained model estimates of τ^2, into the LR statistic (2). Suppose therefore that $\hat{\tau}^2$ is a τ^2 estimator used in the IV test, with $T_{IV}(\hat{\tau}^2)$ denoting the resulting test statistic. When $\hat{\tau}^2$ is substituted for $\hat{\tau}^2_{MLE}$ and $\hat{\tau}^2_{MLE,R}$ in the LR statistic, the resulting MLE for μ is the inverse-variance weighted average and links the two test statistics:

$$\hat{\mu}_{MLE}(\hat{\tau}^2) = \sum_i w_i Y_i \Big/ \sum_i w_i = T_{IV}(\hat{\tau}^2) \Big/ \sqrt{\sum_i w_i},$$
$$w_i = 1/(\hat{\sigma}_i^2 + \hat{\tau}^2).$$

Let $T_{LR}(\hat{\tau}^2) = 2(l(\hat{\mu}_{MLE}(\hat{\tau}^2), \hat{\tau}^2) - l(\mu_0, \hat{\tau}^2))$ denote the resutling LR test statistic. Then we

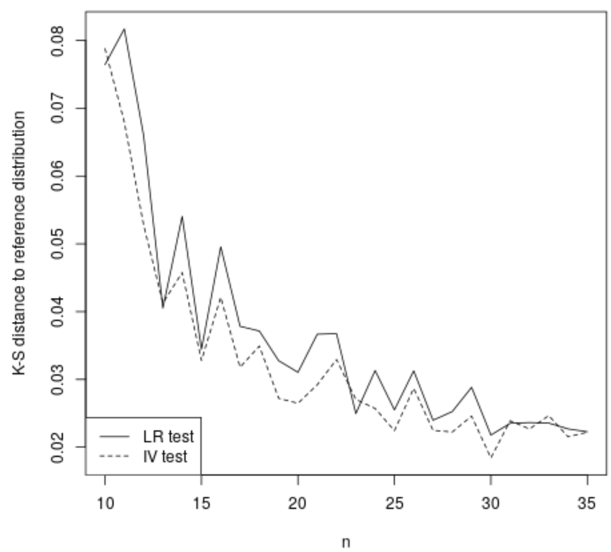


Figure 1: The Kolmogorov-Smirnov distance between empirical CDFs of the LR and IV tests and their reference distributions, chi-squared in the case of the LR test and the standard normal in the case of the IV test, as the sample size n grows. Count data with the log-odds effect estimate.

find its gap with the IV test statistic vanishes,

$$\begin{aligned}
&T_{IV}(\hat{\tau}^2)^2 - T_{LR}(\hat{\mu}_{MLE}(\hat{\tau}^2), \hat{\tau}^2, \hat{\tau}^2) \\
&= T_{IV}(\hat{\tau}^2)^2 + \sum_{i=1}^{n} w_i((Y_i - \mu_0) - (\sum_{i=1}^{n} w_i)^{-1/2} T_{IV}(\hat{\tau}^2))^2 - \sum_{bi=1}^{n} w_i(Y_i - \mu_0)^2 \\
&= T_{IV}(\hat{\tau}^2)^2 - 2(\sum_{i=1}^{n} w_i)^{-1/2}(\sum_{i=1}^{n} w_i(Y_i - \mu_0))T_{IV}(\hat{\tau}^2) \\
&= 2T_{IV}(\hat{\tau}^2)^2 - 2T_{IV}(\hat{\tau}^2)^2 = 0.
\end{aligned}$$

Therefore the difference between the two statistics comes down to the error

$$T_{LR}(\hat{\mu}_{MLE}(\hat{\tau}^2), \hat{\tau}^2, \hat{\tau}^2) - T_{LR}(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}), \hat{\tau}^2_{MLE}, \hat{\tau}^2_{MLE})$$

introduced when $\hat{\tau}^2$ was plugged in for the full and constrained τ^2 MLEs. We expect the discrepancy to come from two places. First, using the same τ^2 estimator under the full and constrained models amounts to estimation with or without knowledge of μ, and under the null is usually valid only when the parameters (μ, τ^2) are orthogonal. Though the parameters in (1) are not orthogonal, the heteroscedasticity arising from the $\hat{\sigma}_i^2$ terms is the only obstacle. Second, the nominal null μ_0 may differ from the the mean of Y, i.e., the alternative hypothesis holds. Proposition 1 formalizes the calculation of the gap with mild regularity assumptions on the data.

Proposition 1. *Let $\hat{\tau}^2 = \hat{\tau}^2(n)$ be a sequence of strictly positive random variables. Assume:*

1. $\frac{d}{dt}\frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(t), t)|_{t=\hat{\tau}^2_{MLE,R}}$ *is bounded away from* 0
2. *Third order derivatives of the likelihood are* $O_P(n)$
3. *The first 2 derivatives of* $\tau^2 \mapsto \hat{\mu}_{MLE}(\tau^2)$ *at* $\tau = \hat{\tau}^2_{MLE}$ *are* $O_P(1)$

Then to leading order as $|\hat{\tau}^2 - \hat{\tau}^2_{MLE}| \to 0$, $|\hat{\tau}^2 - \hat{\tau}^2_{MLE,R}| \to 0$,

$$T_{IV}(\hat{\tau}^2)^2 - T_{LR} = O_P(n) \cdot ((\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2 + (\hat{\tau}^2 - \hat{\tau}^2_{MLE,R})^2). \tag{3}$$

Let $w_i = (\sigma_i + \hat{\tau}^2_{MLE,R})^{-1}$, $i = 1, \ldots, n$. *Then* (3) *may be re-expressed as*

$$T_{IV}(\hat{\tau}^2)^2 - T_{LR} = O_P(n) \cdot ((\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2 + O_P(\sum_i \frac{w_i}{\sum_j w_j}(Y_i - \mu_0))^4 + O_P(\sum_i \frac{w_i^2}{\sum_j w_j^2}(Y_i - \mu_0))^4). \tag{4}$$

Equation (4) decomposes the gap between the tests into two parts. The first is the gap between the τ^2 estimator used in the IV test and the MLE for τ^2. For many common τ^2 estimators this gap is of second order, as the below examples show. The second gap, given by the weighted means in (4), obliquely references the distance between the data and the null μ_0. The null will be defined more carefully in Section 2.2. In the usual RE model the Y_i are indepndent and identically distributed as $N(\mu_0, \tau^2 + \hat{\sigma}^2)$. Then these weighted averages will vanish at a $1/\sqrt{n}$ rate, so that their contribution to $T_{IV}(\hat{\tau}^2)^2 - T_{LR}$ is second order at $Op(1/n)$. However, Proposition 2 below shows that the weighted averages vanish at $1/\sqrt{n}$ rate under more general conditions, allowing e.g., heterogeneous Y_i, finite-sample biases away from the null, and other characteristics of real-world meta-analysis.

We next give examples of $\hat{\tau}^2 - \hat{\tau}^2_{MLE}$, and consequently $T_{IV}(\hat{\tau}^2)^2 - T_{LR}$, for three choices of $\hat{\tau}^2$: restricted maximum likelihood, DerSimonian-Laird, and a naive τ^2 estimator given by the sample variance of the Y_i. We will find that the gap between $\hat{\tau}^2_{MLE}$ and the REML estimator depends only on the number of studies n, while its gap with the sample variance depends only on the sizes of the primary studies, summarized below by a quantity denoted m. On the other hand, we find that the D-L estimator need not converge to the MLE, due ultimately to the inconsistency of the former. Convergence of $\hat{\tau}^2 - \hat{\tau}^2_{MLE}$ leads to conditions for $O(1/n)$ convergence between the LR statistic and the IV statistic using REML or the sample variance.

2.1.1 Restricted maximum likelihood

The objective function for the restricted maximum likelihood estimate of τ^2 is the log likelihood (1) augmented by a penalty term intended to account for the estimation of μ:

$$l_{REML}(\mu(\tau^2), \tau^2) = l(\mu(\tau^2), \tau^2) + C(\tau^2)$$
$$C(\tau^2) = -\frac{1}{2}\log(\sum_{i=1}^{n} \frac{1}{\tau^2 + \sigma_i^2}).$$

The maximum likelihood and restricted maximum likelihood estimates for τ^2 thus satisfy the score equations

$$\frac{dl}{d\tau^2}(\mu(\hat{\tau}^2_{REML}), \hat{\tau}^2_{REML}) + \frac{dC}{d\tau^2}(\hat{\tau}^2_{REML}) = 0$$
$$\frac{dl}{d\tau^2}(\mu(\hat{\tau}^2_{MLE}), \hat{\tau}^2_{MLE}) = 0.$$

Expanding $\hat{\tau}^2_{REML}$ about $\hat{\tau}^2_{MLE}$ in the first score equation and substiuting the second gives, to leading order,

$$\hat{\tau}^2_{REML} - \hat{\tau}^2_{MLE} = -\left(\frac{d^2 l}{d(\tau^2)^2}(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}), \hat{\tau}^2_{MLE}) + \frac{d^2 C}{d(\tau^2)^2}(\hat{\tau}^2_{MLE})\right)^{-1} \frac{dC}{d\tau^2}(\hat{\tau}^2_{MLE}).$$

Since the information $d^2 l/d(\tau^2)^2$ is $O(n)$ and the penalty term C have $O_P(1)$ derivatives, $\hat{\tau}^2_{REML} - \hat{\tau}^2_{MLE} = O_P(1/n)$. Therefore in the proximity of the null hypothesis,

$$T_{IV}(\hat{\tau}^2_{REML})^2 - T_{LR} = O_P(1/n),$$

which is the gap observed in Fig. 1.

2.1.2 Naive estimator

Suppose instead of using the DerSimonian-Laird, REML, or other popular estimator of τ^2, one simply used the sample variance of the effects Y_i, whether with the mean estimated $s^2 = \frac{1}{n}\sum_{i=1}^n (Y_i - \overline{Y})^2$ or taken to be the nominal null $s_0^2 = \frac{1}{n}\sum_{i=1}^n (Y_i - \mu_0)^2$. To analyze the consequences of these choices for $\hat{\tau}^2$ we initially make the simplifcation that the primary studies are all the same size $m = m(n)$, possibly depending on n, and any two-arm studies are balanced. Change variables from $\hat{\sigma}_i^2$ to $V = m \cdot \hat{\sigma}_i^2$. The log-likelihood may be written in terms of V and m as

$$l(\mu, \tau^2) = -\frac{1}{2}\sum_{i=1}^n \log(2\pi(V/m + \tau^2)) - \frac{1}{2}\sum_{i=1}^n \frac{(Y_i - \mu)^2}{V/m + \tau^2}. \tag{5}$$

Then s^2 and $P_n(Y^2)$ are the full and constrained model MLEs when $1/m \to 0$ and (5) becomes the simple homoscedastic Gaussian likelihood. Let $\hat{\tau}^2_{MLE}(m)$ and $\hat{\tau}^2_{MLE,R}(m)$ denote the MLEs of τ^2 under (5) with m viewed as variable. Expanding the score equations with respect to $1/m$ about 0 yields estimates

$$\hat{\tau}^2_{MLE}(m) - s^2 = \frac{1}{ms^2}(\overline{V} - \frac{1}{ns^2}\sum_{i=1}^n V_i(Y_i - \overline{Y})^2) + o(1/m)$$

and

$$\hat{\tau}^2_{MLE,R}(m) - s_0^2 = -\frac{1}{m}\frac{\overline{V}/(2\overline{Y^2}) - (\overline{Y^2})^{-3}\frac{1}{n}\sum_{i=1}^n V_i(Y_i - \mu_0)^2}{1/(2(\overline{Y^2})^2) - (\overline{Y^2})^{-3}\frac{1}{n}\sum_{i=1}^n (Y_i - \mu_0)^2} + o(1/m).$$

Both of these gaps are $O_P(1/m)$ as long the sample moments of the data are bounded. Moreover, if the null hypothesis holds in the generous sense that $P_n(Y) - \mu_0 = O_P(1/\sqrt{n})$, then the gap between the two naive estimators is $|s_0^2 - s^2| = (P_n(Y) - \mu_0)^2 = O_P(1/n)$. Proposition 1 then implies both

$$T_{IV}(s^2)^2 - T_{LR} = O(n/m^2)$$

and

$$T_{IV}(s_0^2)^2 - T_{LR} = O(n/m^2).$$

If in addition the sizes of the primary studies are at least of the same order as the number of studies, $n = O(m)$, these gaps are of second order in n.

2.1.3 DerSimonian-Laird

Suppose as in 2.1.2 that the data are based on studies with a fixed sample size $m = m(n)$ that grows with n and the change of variables $\hat{\sigma}^2 \mapsto V_i/m$ has been carried out; this assumption will be relaxed in Section 2.2. In terms of $(Y_i, V_i), i = 1, \ldots, n$, and m, the DerSimonian-Laird estimator of τ^2 takes the form

$$\hat{\tau}^2_{DL} = \max(0, \frac{P_n(Y^2/V)P_n(1/V) - P_n(Y/V)^2 - P_n(1/V)\frac{1}{m}(1 - \frac{1}{n})}{P_n(1/V)^2 - P_n(1/V^2)\frac{1}{n}}.$$

If the estimator converges to a strictly positive probability limit, a first order approximation ignores the max and

$$\hat{\tau}^2_{DL} = \frac{P_n(Y^2/V)P_n(1/V) - P_n(Y/V)^2}{P_n(1/V)^2} + O_P(1/m) + O_P(1/n).$$

For this expression to congerge to the same probability limit as $s^2 = P_n(Y^2 - P_n(Y))$ requires in general $P_n(Y^2/V) - P_n(Y^2)P_n(1/V) \to_p 0$ and $P_n(Y/V) - P_n(Y)P_n(1/V) \to_p 0$. Since it was shown in 2.1.2 that $\hat{\tau}^2_{MLE}$ is within $O_P(1/m)$ of s^2, $\hat{\tau}^2_{DL}$ and $\hat{\tau}^2_{MLE}$ need not even have the same probability limit. The difference between the $\hat{\tau}^2_{DL}$ and s^2 examples is that, in the absence of the uncorrelatedness conditions just mentioned, $\hat{\tau}^2_{DL}$ need not be a consistent estimator for large m and n, even under the usual strict definition of the null that $\mathbb{E}Y_i = \mu_0$. As a result the likelihood ratio test and IV estimator based on $\hat{\tau}^2_{DL}$ can behave very differently. See Section 3.2, showing a range of settings where the LR test is consistent but the IV test using D-L is not.

2.2 Asymptotic regime and statistical model

The examples given in 2.1.2 and 2.1.3 show how the sample sizes of the primary studies arise naturally in formulating results about the performance of meta-analysis tests. At the same time, a realistic theoretical account of meta-analysis acknowledges that, even if number of studies n grows arbitrarily large, the reported effects and standard errors remain estimates limited in accuracy by the study sizes. We therefore introduce a model of meta-analysis taking into account both asymptotic considerations, the number of studies and the sizes of the studies. We avoid making strong distributional assumption and will mostly rely on moment assumptions, while attempting to capture the basic features of a random effects model.

The usual random effects model is that, conditionally on the standard errors $\hat{\sigma}_1^2, \ldots, \hat{\sigma}_{n,}^2$, the Y_i are independently distributed as $N(\mu_0, \tau_0^2 + \hat{\sigma}_i^2)$. While convenient and plausible, this model is in practice the source of misspecifiction errors, some of which are discussed below. We therefore generalize it. A basic characteristic of the random effect model is to divide the variance of the observed effects Y_i into a main component and a sampling error

$$\begin{aligned} \mathrm{Var} Y_i &= \text{main term} + \text{ sampling error} \\ &= \tau_i^2 + \sigma_i^2, \end{aligned}$$

say. While the main term is generally $O(1)$ as the number of studies or their sizes grow, the sampling errors generally vanishes with the sample size of the studies. We therefore assume that the variances can be divided in this way. We define the heterogeneity parameter as

$$\tau_0^2 = \frac{1}{n} \sum_{i=1}^{n} \tau_i^2$$

and introduce an additional asymptotic parameter m defined through

$$\frac{1}{m} = \frac{1}{n} \sum_{i=1}^{n} \sigma_i^2.$$

The average variance of independent Y_i may then be written

$$Var_n(Y) = \tau_0^2 + O(\frac{1}{m}).$$

We stipulate that

$$\inf_{n,m} Var_n(Y) > 0 \tag{6}$$

so that τ_0^2 is identifiable as the limiting variance as the number of studies and their sample sizes grow. This definition of τ_0^2 agrees with the usual definition derived under a random effects model, but without positing a random effect, and allowing for heterogeneity of the Y_i distributions. Furthermore, since important statistical properties of $\hat{\sigma}_i^2$ are difficult to study when $\hat{\sigma}_i^2$ are vanishing, we change variable to

$$(Y_i, \hat{\sigma}_i^2) \mapsto (Y_i, V_i), \text{ where } V_i = m\hat{\sigma}_i^2.$$

In this way we can study the behavior of the test statistics as $n \to \infty, m \to \infty$ with the variables Y and V remaining $O_P(1)$. This flexibility is needed outside the idealized Guassian model, where Y_i and V_i are often correlated. For example the SMD and its variance estimate are functionally related.

Our model for the null hypothesis that the Y_i are located at μ_0 is

$$E_n(Y) - \mu_0 = O(\frac{1}{m}) \tag{7}$$

The allowance of an $O(\frac{1}{m})$ error contrasts with the precise null $\mathbb{E}Y_i = 0, i = 1, \ldots, n$, of the normal-normal model. Many effect estimators like the log-odds and SMD exhibit an $O(\frac{1}{m})$ bias away from the null. A consequence of this model is that, provided the Y_i have bounded variances,

$$\overline{Y} - \mu_0 = (\overline{Y} - E_n(Y)) + (E_n(Y) - \mu_0) = O_P(\frac{1}{\sqrt{n}}) + O(\frac{1}{m}). \tag{8}$$

Finally, since we state results for a generic estimator of τ_0^2, we assume a similar representation for $\hat{\tau}^2$,

$$\hat{\tau}^2 - \tau_0^2 = \overline{Z} + O_P(1/m) \text{ where } \overline{Z} = O_P(\frac{1}{\sqrt{n}}). \tag{9}$$

We use "null model" to refer to the modeling assumptions (6), (7), and (9), as well as the presupposed definitions of m, τ_0^2, and V. Section 2.3 studies the consistency and rates of the IV test, and the LR test as well via Proposition 1, under the proposed model for the null hypothesis.

2.3 Consistency and performance of the tests under the null

In this section we first show that under the null model of Section 2.2 the contributions of the weighted means in Proposition 1 is of second order, implying in particular that the difference between the IV test with REML and the LR test is $O(1/n)$. Then we establish conditions for consistency for the two tests. Finally we give conditions under which the coverage of confidence intervals converge to their nominal level at an $O(1/n)$ rate.

Proposition 2 shows that when the null model of Section 2.2 holds and the order of the primary study sizes is at least $1/\sqrt{n}$, the weighted means in (4) vanish at an $O_P(1/\sqrt{n})$ rate. As a consequence the gap between the IV and LR tests is determined by the choice of $\hat{\tau}^2$ in the IV test up to second order.

Proposition 2. *Assume:*

1. $\sup_n E_n(Y^4) < \infty$
2. *the null model bias assumption* (7) *and non-degeneracy assumption* (6)
3. $P_n(V) = O_P(1)$, $P_n(V^2) = O_P(1)$
4. $n = O(m^2)$

Then

$$T_{IV}(\hat{\tau}^2)^2 - T_{LR} = O_P(n) \cdot (\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2 + O_P(1/n).$$

Proposition 2 allows asymptotic results for both the LR test and IV test to be obtained by establishing the result for just one of the two tests, at least when the null model of Section 2.2 holds. We first give conditions for consistency of the tests.

Proposition 3. *Let $T_{IV}(\hat{\tau}^2)$ denote the IV test statsitic using $\hat{\tau}^2$ as estimator for τ^2. Under the assumptions of Lemma 7:*

1. *As $n \to \infty$ the leading order bias of $T_{IV}(\hat{\tau}^2)$ is*

$$\frac{\sqrt{n}}{\tau}(E_n(Y) - \mu_0) - \frac{\sqrt{n}P_n(YV)}{m\tau^3}. \tag{10}$$

2. *Assume further:*

   (a) Lindeberg condition for Y_i

   (b) $m = o(n^2)$

   Then $T_{IV}(\hat{\tau}^2)$ converges to a standard normal random variable.

The bias term (10) may arise either due to mean bias away from the null μ_0 in the effect estimates Y_i or correlation between the effect estimatess and their standard errors. Both of these are present in common estimators such as the log-odds or SMDs, absent special conditions like perfectly balanced arms. When present, consistency of the tests requires $n = o(m^2)$. However, as the bias and correlation terms are in general $O(\sqrt{n}/m)$, the rate at which the test converges can be arbitrarily slow as the order of m approaches $\sqrt{n}$ from above. This lower limit at $\sqrt{n}$ for consistency is unsurprising since if $n \neq o(m^2)$ then Y_i and $\hat{\tau}^2$ can themselves be asymptotically biased. That is, an effect estimate like the log-odds need not be centered at μ_0 when the null $\mu = \mu_0$ holds even in the limit as $n \to \infty, m \to \infty$ unless $n = o(m^2)$, so tests of $\mu = \mu_0$ based on them can be expected to be inconsistent.

Proposition 4 presents an Edgeworth expansion for $T_{IV}(\hat{\tau}^2)$, which quantifies rate of convergence of its CDF to the Gaussian CDF up to second order error.

Proposition 4. *Assuming:*

1. *null model assumption* (8) *and $\hat{\tau}^2$ influence function* (9)
2. *(Y, V, Z) have mixed moments up to order 3*
3. *$n = O(m)$*

then

$$\begin{aligned}
&\mathbb{P}(T_{IV}(\hat{\tau}^2) \leq x) \\
&= \Phi(x) + \frac{\kappa_3}{6\kappa_2^{3/2}}(1 - x^2)\phi(x) - \kappa_1(1 + \frac{\kappa_3}{6\kappa_2^{3/2}}(x^3 - 3x))\phi(x) + O_P(\frac{1}{m}) + O_P(\frac{1}{n})
\end{aligned}$$

uniformly in x, where

$$\begin{aligned}
\kappa_1 &= \kappa_1(T_{IV}(\hat{\tau}^2)) = \frac{\sqrt{n}}{\tau_0}E_n(Y - \mu_0) - \frac{1}{\tau_0^3}(\frac{1}{2\sqrt{n}}Cov_n(Y, Z) + \frac{\sqrt{n}}{m}Cov_n(Y, V)), \\
\kappa_3 &= \kappa_3(T_{IV}(\hat{\tau}^2)) = n^{-1/2}E_n((Y - E_n(Y))^3)/\tau_0^3
\end{aligned}$$

are the first and third cumulants of $T_{IV}(\hat{\tau}^2)$ up to error $O_P(1/m) + O_P(1/n)$.

Under the assumptions of Proposition 4, in particular that the primary studies have the same or greater order than the number of studies, the two-sided IV test and the LR test achieve $O_P(1/n)$ rates.

Corollary 5. *Under the assumptions of Proposition 4*

$$\mathbb{P}(|T_{IV}(\hat{\tau}^2)| \leq z_{1-\alpha/2}) = 1 - \alpha + O_P(1/n).$$

In Proposition 4 the assumption that $n = O(m)$ was imposed, rather than $O(m^2)$ as in Propositions 2 and 3, mainly to facilitate certain calculations in the proof. While Proposition 3 shows that the tests are in general inconsistent when $n \neq o(m^2)$, one might wonder if the $1/n$ rate could be achieved for m in the range $\sqrt{n}$ to n. Simulations 3.2 suggest the rate actually interpolates linearly between these boundaries.

3 Simulation

In this section we study the results of Section 2 using synthetic data, mainly for confirmation of the theory but also to see how robust the conclusions are as assumptions cease to be met. Section 3.1 looks specifically at the results of Section 2.1 relating the IV and LR tests to each other. In focusing on the gap between the statistics, these simulations leave open the question of the actual performance of the tests. The test rates themselves are studied in 3.2.

All the simulations follow a similar design, reflecting our focus on double asymptotics. In this design, the convergence rate of a quantity is studied as the study size m varies between $n^{1/5}, \ldots, n^2$. For fixed m, the convergence rate of the quantity under study is estimated as the slope coefficient in a log-log regression of the statistic versus n, as $n = 10, \ldots, 25$.

Other than count data simulations in Figs. 1 and 3b, the simulations use the following data-generating process. The pairs $(Y_i, V_i), i = 1, \ldots, n$, are generated as independent and identically distributed bivariate gamma random vectors,

$$\begin{aligned} Y &\sim X_1 + X_3 \\ V &\sim X_2 + X_3 \\ X_i &\sim \Gamma(a_i, b), i = 1, 2, 3. \end{aligned} \tag{11}$$

The shape parameters a_i and common rate b are chosen so that $\mathbb{E}Y = \mu_0 + 1/m$, $\text{Var}Y = \tau_0^2 + 1/m$, $\text{Var}V = 1$, and $\text{corr}(Y, V) = .4$. This adversarial data generating process is chosen to ensure that the results don't depend on the nice properties of Gaussian data, such as symmetry and independence of the sample mean and variance, while allowing the first and second moment structure to be directly specified. Also provided is a simulation using more realistic count data. For data with sample size n, $B = n^3$ monte carlo repetitions are used, so that convergence rates up to $O(1/n)$ may be observed.

3.1 Test equivalence

Fig. 2a presents a simulation examining $|\hat{\tau}^2 - \hat{\tau}^2_{MLE}|$. According to Proposition 1 this quantity drives the gap between the the LR test and an IV test based on $\hat{\tau}^2$. As expected from the analysis Section 2.1, the REML estimator converges to the MLE at a $1/n$ rate irrespective of m, the naive estimator converges at a $1/m$ rate irrespective of n, and the D-L estimator does not converge to the MLE when $\mathrm{corr}(Y, V)$ is nonzero. The Hunter-Schmidt estimator behaves similarly to the D-L. The other estimators are intermediate, converging at a $1/m$ rate as m varies between $\sqrt{n}$ and n at which point the $1/n$ floor on the rate comes into effect.

Fig. 2b presents a simulation examining the gap between the test statistics themselves at the null. According to Proposition 2, the gap $|\hat{\tau}^2 - \hat{\tau}^2_{MLE}|$ differs from the gap $|T_{IV}(\hat{\tau}^2)^2 - T_{LR}|$ by a factor of $O(n)$. We therefore expect, and the figure confirms, that 1. $|T_{IV}(\hat{\tau}^2_{REML})^2 - T_{LR}| = O_P(1/n)$, 2. $|T_{IV}(\hat{\tau}^2)^2 - T_{LR}| = O_P(1/n) + O_P(1/m)$ for consistent $\hat{\tau}^2$, and 3. the DerSimonian-Laird estimator fails to converge.

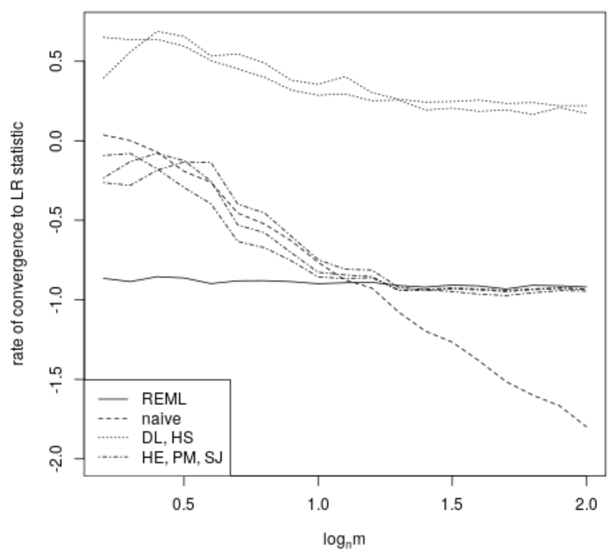
3.2 Test rates

Fig. 3a presents a simulation examining the rates of convergence of the p-values of the tests to their nominal values. Specifically, the Kolmogorov-Smirnov distance is obtained as the maximum distance betwen the empirical CDF of a sample of size B of the test statistic to a chi-squared distribution, in the case the test statistic is the LR statistic, or the Gaussian distribution, in the case of the IV test statistics. The rates that these K-S distances vanish are then estimated. As expected from Corollary 5, all the consistent tests perform similarly. They fail to be consistent for $\log_n m \leq \frac{1}{2}$ and achieve a $1/n$ rate for $\log_n m \geq 1$, with the intermediate rates appearing to be linearly interpolated.

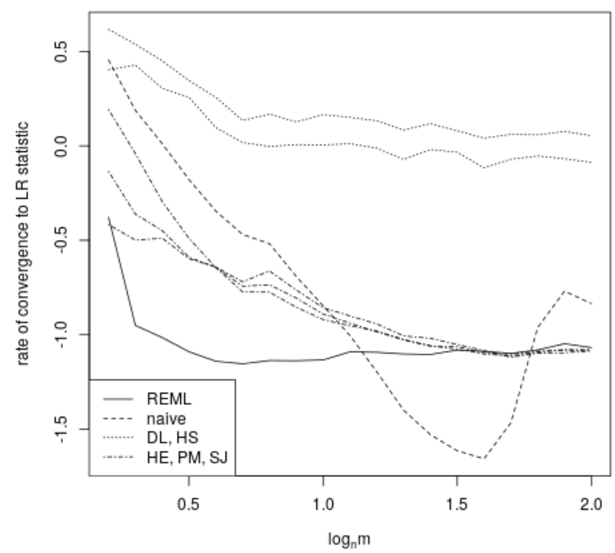
Finally, Fig. 3b repeats the design of Fig. 3a using more realistic data. Rather than the bivariate gamma model (11), count data on two-armed trials are used. First n random effects B_i, representing the log-odds effect underlying study i, are sampled from a centered exponential distribution. The exponential distribution is used rather than the more common normal distribution to ensure as above that Gaussianity of the random effect isn't necessary for the results. Next, p_C, representing the event rate in the control group, is sampled from a beta distribution, which together with B_i implies a treatment event rate p_T. Cell counts x_C and x_T are then sampled as binomial with probabilities p_C and p_T, with $m_C = m$ and $m_T = 1.5m_C$ trials, a $2:3$ imbalance across arms. The summary data are then obtained as $Y_i = \log \frac{p_C(1-p_T)}{(1-p_c)p_T}$ and $\hat{\sigma}_i^2 = \frac{1}{x_C} + \frac{1}{m_C - x_C} + \frac{1}{x_T} + \frac{1}{m_T - x_T}$. The results are consistent with those of Fig. 3a, where the bivariate gamma data was used.

4 Discussion

We have shown that when the null hypothesis holds even approximately, the inverse-variance weighted test using the REML estimator agrees with the likelihood ratio test up to error $O(1/n)$. For estimators besides REML, the typical rate is $O(1/n) + O(1/m)$, where m summarizes the sizes of the primary studies. In showing that many of these IV variants are second-order equivalent to the LR test, we of course show that they are equivalent to each other, with exceptions like D-L. It is to be expected that two consistent τ^2 estimators are with $O_P(1/\sqrt{n})$ of each other, and that two IV tests based on these two estimators would exhibit the same gap. Perhaps more surprising is that the τ^2 estimators and resulting IV tests appear to agree up to second order $O_P(1/n) + O_P(1/m)$.

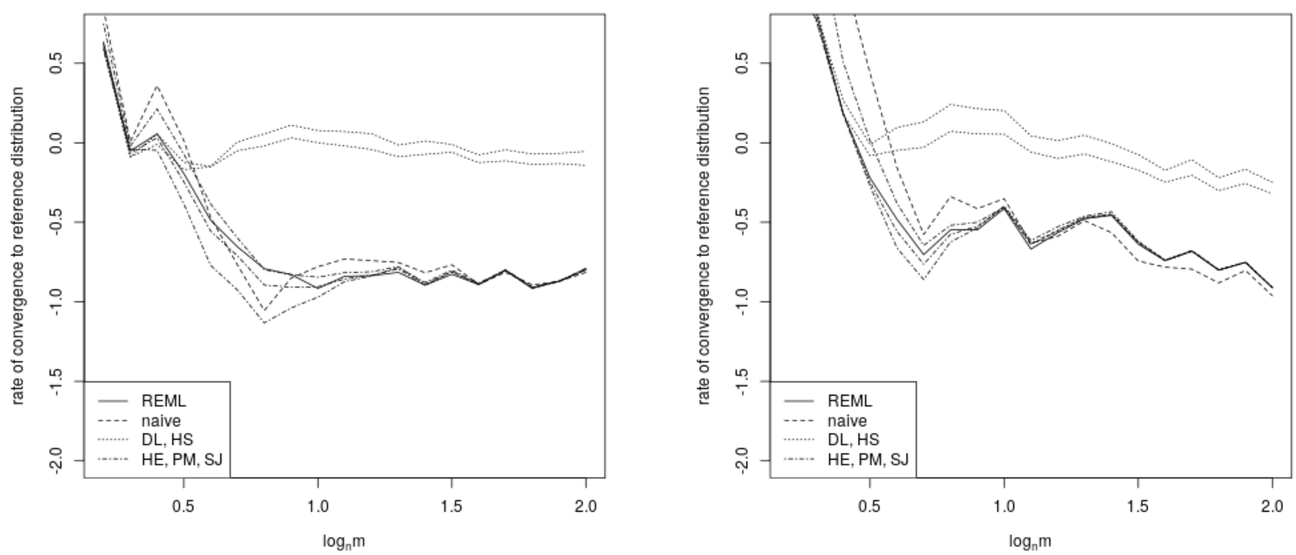


(a) The rate of convergence of $\hat{\tau}^2$ to $\hat{\tau}^2_{MLE}$. When $\hat{\tau}^2$ is REML the rate is $O(1/n)$ irrespective of m. When $\hat{\tau}^2$ is the sample variance the rate is $O(1/m)$ irrespective of n. The D-L and H-S estimators (grouped) don't converge to the MLEs for this data, due to correlation between Y and V. The H-E, P-M, and S-J estimators (grouped) converge at an $O(1/n) + O(1/m)$ rate.



(b) The rate of convergence of $T_{IV}(\hat{\tau}^2)^2$ to T_{LR}. With the exception of teh D-L and H-S estimators (grouped), the convergence rate between the IV and LR tests drops to $O(1/n)$ as $m \uparrow n$, reflecting an $O(1/n)$+$O(1/m)$ rate. When $\hat{\tau}^2$ is REML the $O(1/n)$ convergence rate is observed across m values, slowing only as m approaches constancy.

Figure 2: Simulations on the equivalence of the LR and IV tests (Section 2.1). Bivariate gamma data.



(a) Bivariate gamma data.

(b) Count data with the log-odds ratio estimator.

Figure 3: Simulations on the consistency and convergence rates of the LR and IV tests (Section 2.3). The rate of convergence of $T_{IV}(\hat{\tau}^2)^2$ and T_{LR} to their nominal reference distributions is plotted against the study sizes m.

We have also shown that under relatively general conditions the two tests are inconsistent when m isn't at least of order $\sqrt{n}$, while they achieve $1/n$ rates when m is of order n or greater.

An important limitation is that we have focused on data at or near the null hypothesis. It is plausible that the inconsistency of the DerSimonian-Laird estimator noted above is counterbalanced by better power. A second limitation is that we have not studied behavior near the boundary of the random effect model, as $\tau^2 \to 0$. Third, we have treated the many $\hat{\tau}^2$ variants of the IV test in a basic manner. For example, we always referred the IV test statistic to a normal distribution, whereas Student's t is sometimes advocated (Follmann and Proschan, 1999). Fourth, we have considered the effect estimators in a very basic manner. For example, the log-odds ratio and standardized mean difference both have de-biased variants that obviate some of the modeling considerations of Section 2.2.

The software used to carry out the simulations and prepare the figures is publicly available at the author's website.

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Proof of Proposition 1. Let $\hat{\theta}_{MLE}$ and $\hat{\theta}(\hat{\tau}^2)$ denote the vectors $(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}), \hat{\tau}^2_{MLE}, \hat{\tau}^2_{MLE,R})$ and $(\hat{\mu}_{MLE}(\hat{\tau}^2), \hat{\tau}^2, \hat{\tau}^2)$ and treat T_{LR} as a function of the 3 MLEs, so that in this notation the usual LR statistic is written $T_{LR}(\hat{\theta}_{MLE})$. By the remarks preceding the proposition statement,

$$T_{IV}(\hat{\tau}^2)^2 - T_{LR}(\hat{\theta}_{MLE}) = T_{LR}(\hat{\theta}(\hat{\tau}^2)) - T_{LR}(\hat{\theta}_{MLE}).$$

Without assuming $\hat{\theta}(\hat{\tau}^2)$ and $\hat{\theta}_{MLE}$ are close, a formal Taylor expansion with respect to $\hat{\theta}(\hat{\tau}^2)$ about $\hat{\theta}_{MLE}$ is

$$\begin{aligned} &T_{LR}(\hat{\theta}(\hat{\tau}^2)) - T_{LR}(\hat{\theta}_{MLE}) \\ &\qquad = \frac{1}{2}(\hat{\theta}(\hat{\tau}^2) - \hat{\theta}_{MLE})T''_{LR}(\hat{\theta}_{MLE})(\hat{\theta}(\hat{\tau}^2) - \hat{\theta}_{MLE}) + o((\hat{\theta}(\hat{\tau}^2) - \hat{\theta}_{MLE})^2)O_P(n) \end{aligned} \tag{12}$$

where the linear term vanishes by the score equations for the model and the assumed bound on likelihood third derivatives was used to control the error. The matrix $T''_{LR}(\hat{\theta}_{MLE})$ is the Hessian of $(\mu, \tau^2, \tau_R^2) \mapsto 2(l(\mu, \tau^2) - l(\mu_0, \tau_R^2))$ at the MLEs $(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}, \hat{\tau}^2_{MLE,R})$ and is block-diagonal with blocks given by the full and constrained information matrices,

$$T''_{LR}(\hat{\theta}_{MLE}) = 2\begin{pmatrix} \frac{d^2l}{d\mu^2}(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}) & \frac{d^2l}{d\mu d\tau^2}(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}) & 0 \\ \frac{d^2l}{d\tau^2 d\mu}(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}) & \frac{d^2l}{d(\tau^2)^2}(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}) & 0 \\ 0 & 0 & -\frac{d^2l}{d(\tau^2)^2}(\mu_0, \hat{\tau}^2_{MLE,R}) \end{pmatrix}.$$

These matrix entries are in general $O_P(n)$ so the gap (12) is

$$\begin{aligned} &T_{LR}(\hat{\theta}(\hat{\tau}^2)) - T_{LR}(\hat{\theta}_{MLE}) \\ &= O_P(n)((\hat{\mu}_{MLE}(\hat{\tau}^2) - \hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}))^2 + 2(\hat{\mu}_{MLE}(\hat{\tau}^2) - \hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}))(\hat{\tau}^2 - \hat{\tau}^2_{MLE}) + (\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2 \\ &- (\hat{\tau}^2 - \hat{\tau}^2_{MLE,R})^2) + o((\hat{\theta}(\hat{\tau}^2) - \hat{\theta}_{MLE})^2)O_P(n). \end{aligned}$$

The $\hat{\mu}_{MLE}$ terms in may be absorbed into the $\hat{\tau}^2_{MLE}$ terms by

$$\begin{aligned} \hat{\mu}_{MLE}(\hat{\tau}^2) - \hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}) &= \sum_i \left(\frac{w_i(\hat{\tau}^2)}{\sum_i w_j(\hat{\tau}^2)} - \frac{w_i(\hat{\tau}^2_{MLE})}{\sum_i w_j(\hat{\tau}^2_{MLE})}\right)Y_i \\ &= (\hat{\tau}^2 - \hat{\tau}^2_{MLE})\frac{d}{d\tau^2}\sum_i \left(\frac{w_i(\tau^2)}{\sum_i w_j(\tau^2)}Y_i\right)\Bigg|_{\tau^2 = \hat{\tau}^2_{MLE}} + o((\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2) \\ &= (\hat{\tau}^2 - \hat{\tau}^2_{MLE}) \cdot O_P(1). \end{aligned}$$

The assumed bounds on the derivatives of $\hat{\mu}_{MLE}$ were used to control the error. Therefore, to leading order,

$$T_{IV}(\hat{\tau}^2)^2 - T_{LR} = O_P(n) \cdot ((\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2 + (\hat{\tau}^2 - \hat{\tau}^2_{MLE,R})^2),$$

which implies

$$T_{IV}(\hat{\tau}^2)^2 - T_{LR} = O_P(n) \cdot ((\hat{\tau}^2 - \hat{\tau}^2_{MLE})^2 + (\hat{\tau}^2_{MLE} - \hat{\tau}^2_{MLE,R})^2).$$

As a final step we approximate $\hat{\tau}^2_{MLE} - \hat{\tau}^2_{MLE,R}$. Expanding the score equation for the full model τ^2,

$$\begin{aligned}0 &= \frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE}), \hat{\tau}^2_{MLE}) \\ &= \frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE,R}), \hat{\tau}^2_{MLE,R}) + (\hat{\tau}^2_{MLE} - \hat{\tau}^2_{MLE,R})\frac{d}{dt}\frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(t), t)|_{t=\hat{\tau}^2_{MLE,R}} + O_P(|\hat{\tau}^2_{MLE} - \hat{\tau}^2_{MLE,R}|^2).\end{aligned}$$

Rearranging and dropping higher order terms,

$$\hat{\tau}^2_{MLE} - \hat{\tau}^2_{MLE,R} = -(\frac{d}{dt}\frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(t), t)|_{t=\hat{\tau}^2_{MLE,R}})^{-1}\frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE,R}), \hat{\tau}^2_{MLE,R}).$$

The total derivative in the denominator is assumed to be bounded away from 0. Making use of the constrained model score equation

$$0 = \frac{dl}{d\tau^2}(\mu_0, \hat{\tau}^2_{MLE,R}) = -\sum_i w_i + \sum_i w_i^2(Y_i - \mu_0)^2$$

the numerator may be re-expressed as

$$\begin{aligned}2\frac{dl}{d\tau^2}(\hat{\mu}_{MLE}(\hat{\tau}^2_{MLE,R}), \hat{\tau}^2_{MLE,R}) &= -\sum_i w_i + \sum_i w_i^2(Y_i - \hat{\mu}_{MLE}(\hat{\tau}^2_{MLE,R}))^2 \\ &= \sum_i w_i^2((Y_i - \hat{\mu}_{MLE}(\hat{\tau}^2_{MLE,R}))^2 - (Y_i - \mu_0)^2) \\ &= \sum_i w_i^2 \sum_i \frac{w_i}{\sum_j w_j}(Y_i - \mu_0)(\sum_i \frac{w_i}{\sum_j w_j}(Y_i - \mu_0) - 2\sum_i \frac{w_i^2}{\sum_j w_j^2}(Y_i - \mu_0)) \\ &= O_P(n) \cdot O_P(\sum_i \frac{w_i}{\sum_j w_j}(Y_i - \mu_0))^2 + O_P(\sum_i \frac{w_i^2}{\sum_j w_j^2}(Y_i - \mu_0))^2.\end{aligned}$$

□

Lemma 6. *Assume:*

1. $m \to \infty$
2. $\sup_n E_n(Y^4) < \infty$
3. *the non-degeneracy assumption* (6)

Then $\hat{\tau}^2_{MLE,R} - E_n((Y - \mu_0)^2) = O_P(1/\sqrt{n})$.

Proof of Lemma 6. The score equation for $\hat{\tau}^2_{MLE,R}$ is

$$\begin{aligned}0 = \frac{1}{n}\frac{dl}{d\tau^2}(\mu_0, \tau^2) &= \frac{1}{2n}\sum_i \frac{(Y_i - \mu_0)^2}{(\tau^2 + V_i/m)^2} - \frac{1}{2n}\sum_i \frac{1}{\tau^2 + V_i/m} \\ &= \frac{1}{2\tau^2}(\frac{1}{\tau^2 n}\sum_i (Y_i - \mu_0)^2 - 1) + O(1/m). \qquad (13)\end{aligned}$$

The mean of the main term $\frac{1}{2\tau^2}(\frac{1}{\tau^2}E_n((Y-\mu_0)^2)-1)$ vanishes as $n\to\infty$ iff $\tau^2 - E_n((Y-\mu_0)^2)\to 0$, so $\hat\tau^2_{MLE,R} - E_n((Y-\mu_0)^2)\to 0$.

Denote $\tau_*^2 = E_n((Y-\mu_0)^2)$. Expanding the score equation $dl/d(\tau^2)(\mu_0,\hat\tau^2_{MLE,R})=0$ with respect to $\hat\tau^2_{MLE,R}$ about τ_*^2,

$$\hat\tau^2_{MLE,R} - \tau_*^2 = -\frac{\frac{1}{n}\frac{dl}{d\tau^2}(\mu_0,\tau_*^2)}{\frac{1}{n}\frac{d^2l}{d(\tau^2)^2}(\mu_0,\tau_*^2)} + O_P\left(\frac{(\hat\tau^2_{MLE,R}-\tau_*^2)^2}{|\frac{d^2l}{d(\tau^2)^2}(\mu_0,\tau_*^2)|}\right).$$

The denominators are bounded in probability away from 0 as long as $\tau_*^2>0$. The numerator of the main term is given by (13), and therefore $\hat\tau^2_{MLE,R}-\tau_*^2$ and $P_n((Y-\mu_0)^2)-\tau_*^2$ converge at the same rate. By the fourth moment assumption, the latter converges at the parametric rate. □

Proof of Proposition 2. To make the notation more concise in this proof we take $\mu_0=0$. By Prop. 1, the conclusion of this proposition follows on establishing that $\sum_i \frac{w_i}{\sum_j w_j}Y_i$ and $\sum_i \frac{w_i^2}{\sum_j w_j^2}Y_i$ are $O_P(1/\sqrt{n})$. That is, from the assumed parametric convergence rate of the simple mean $P_n(Y)$ we are to deduce the rate for the weighted means. To do so we show that $\sum_{i=1}^n(\frac{w_i}{\sum_j w_j}-\frac{1}{n})Y_i = O_P(1/\sqrt{n})$ and $\sum_{i=1}^n(\frac{w_i^2}{\sum_j w_j^2}-\frac{1}{n})Y_i = O_P(1/\sqrt{n})$.

Let $\epsilon_i = |\frac{w_i}{\sum_j w_j}-\frac{1}{n}|$. Then $\sum_i(\frac{w_i}{\sum_j w_j}-\frac{1}{n})Y_i \le (\sum \epsilon_i^2)^{1/2}(\sum Y_i^2)^{1/2}$ is $O_P(1/\sqrt{n})$ if $\sum\epsilon_i^2 = O_P(1/n^2)$. Let $\delta = |\hat\tau^2_{MLE,R}-\tau_*^2|$; by Lemma 6, $\delta = O_P(1/\sqrt{n})$. For n large enough, $\{0<\tau_*^2-\delta\}$ holds with arbitrary certainty. On this event

$$\frac{1}{\tau_*^2+\delta+\hat\sigma_i^2} \le \frac{1}{\hat\tau^2_{MLE,R}+\hat\sigma_i^2} = w_i \le \frac{1}{\hat\tau^2_{MLE,R}} \le \frac{1}{\tau_*^2-\delta}.$$

Let $a_i = (\tau_*^2+\delta+\hat\sigma_i^2)^{-1}$ and $b=(\tau_*^2-\delta)^{-1}$. Then

$$\begin{aligned}
\frac{a_i}{nb} &\le \frac{w_i}{\sum_j w_j} \le \frac{b}{\sum_j a_j}\\
\epsilon_i &\le \max(\frac{1}{n}|\frac{a_i}{b}-1|, |\frac{b}{\sum a_j}-\frac{1}{n}|),\\
\sum_i \epsilon_i^2 &\le \frac{1}{n^2}\sum_i \max((\frac{a_i}{b}-1)^2, (\frac{b}{\frac{1}{n}\sum a_j}-1)^2)\\
&\le \frac{1}{n^2}(\sum_i(\frac{a_i}{b}-1)^2 + n(\frac{b}{\frac{1}{n}\sum a_j}-1)^2),
\end{aligned} \tag{14}$$

and it suffices to show that $\sum_i(\frac{a_i}{b}-1)^2$ and $n(\frac{b}{\frac{1}{n}\sum a_j}-1)^2$ are $O_P(1)$.

For the first term on the RHS of (14),

$$\begin{aligned}
\sum_i(\frac{a_i}{b}-1)^2 &= \sum_i(\frac{\tau_*^2-\delta}{\tau_*^2+\delta+\hat\sigma_i}-1)^2\\
&\le \frac{1}{\tau_*^2}(n\delta^2+2n\delta P_n(\hat\sigma)^2+nP_n(\hat\sigma)^4)
\end{aligned}$$

is $O_P(1)$ if δ and $P_n(\hat{\sigma}^2)$ are $O_P(1/\sqrt{n})$ and $P_n(\hat{\sigma}^4) = O_P(1/n)$. These follow from our assumptions since $P_n(\hat{\sigma}^2) = 1/mP_n(V)$, $P_n(\hat{\sigma}^4) = 1/m^2 P_n(V)^2$, and $n = O(m^2)$. For the second term on the RHS of (14) it suffices to show $\frac{b}{a_i} - 1 = O_P(1/\sqrt{n})$. By Jensen's inequality,

$$\frac{1}{\tau_*^2 + \delta + P_n(\hat{\sigma})^2} \le P_n(a) \le \frac{1}{\tau_*^2 - \delta}$$
$$1 \le \frac{b}{P_n(a)} \le 1 + \frac{2\delta + P_n(\hat{\sigma}^2)}{\tau_*^2 - \delta}.$$

Since δ and $P_n(\hat{\sigma}^2)$ are $O_P(1/\sqrt{n})$ as above, the result follows. The same method of proof works for the squared weights $w_i^2/\sum_j w_j^2$.

$\square$

Lemma 7. *Assuming*

1. *null model assumption* (8) *and* $\hat{\tau}^2$ *influence function* (9)
2. $P_n(Y)$, $P_n(V)$, $P_n(YV)$ *are* $O_P(1)$, *and* $P_n(V^2)$,$P_n(YV^2)$ *are* $O_P(m)$

Then $T_{IV}(\hat{\tau}^2)$ *has first- and second-order expansions*

$$T_{IV}(\hat{\tau}^2) = \sqrt{n}P_n(Y)(\tau^{-1} - \frac{1}{2}\tau^{-3}P_n(Z)) - \frac{\sqrt{n}}{m}P_n(YV)(\tau^{-3} - \frac{3}{2}\tau^{-5}P_n(Z)) + O_P(\frac{\sqrt{n}}{m^2}) + O_P(\frac{1}{n}) + O_P(\frac{1}{m}) \tag{15}$$

$$= \frac{\sqrt{n}}{\tau}P_n(Y - E_n(Y)) + \frac{\sqrt{n}}{\tau}(E_n(Y) - \mu_0) - \frac{\sqrt{n}P_n(YV)}{m\tau^3} + O_P(\frac{1}{\sqrt{n}}) + O_P(\frac{\sqrt{n}}{m^2}) + O_P(\frac{1}{m}). \tag{16}$$

Proof of Lemma 7. After changing variables $\hat{\sigma}_i^2 \mapsto V_i/m$ and setting $\mu_0 = 0$ the IV test statistic using $\hat{\tau}^2$ is written

$$T_{IV} = \frac{\sum_{i=1}^n w_i Y_i}{\sqrt{\sum_{i=1}^n w_i}}$$
$$w_i = 1/(\hat{\tau}^2 + V_i/m).$$

Expanding in $1/m$ to first order,

$$T_{IV}(\hat{\tau}^2) = \sqrt{\frac{n}{\hat{\tau}^2}}P_n(Y) + \frac{\sqrt{n}}{2m(\hat{\tau}^2)^{3/2}}P_n(Y)P_n(V) - \frac{\sqrt{n}}{m(\hat{\tau}^2)^{3/2}}P_n(YV) + O_P(\frac{1}{m^2})\frac{d^2}{d(1/m)^2}T_{IV}(\hat{\tau}^2)\big|_{1/m\to 0}.$$

The second derivative in the error is

$$\frac{\sqrt{n}}{m}(\frac{P_n(Vw^2)P_n(YVw^2)}{(P_n(w))^{3/2}} + \frac{1}{m}(\frac{P_n(Yw)P_n(Vw^2)}{(P_n(w))^{3/2}} - \frac{2P_n(YVw^2)}{(P_n(w))^{1/2}})$$
$$+ \frac{1}{m^2}(\frac{3P_n(Yw)(P_n(Vw^2))^2}{4(P_n(w))^{5/2}} - \frac{P_n(Yw)P_n(V^2w^3)}{(P_n(w))^{3/2}} + \frac{2P_n(YV^2w^3)}{(P_n(w))^{1/2}}))$$

which is $O_P(\sqrt{n}/m)$ under the empirical moment assumptions. Therefore

$$T_{IV}(\hat{\tau}^2) = \sqrt{\frac{n}{\hat{\tau}^2}} P_n(Y) + \frac{\sqrt{n}}{2m(\hat{\tau}^2)^{3/2}} P_n(Y)P_n(V) - \frac{\sqrt{n}}{m(\hat{\tau}^2)^{3/2}} P_n(YV) + O_P(\frac{\sqrt{n}}{m^2}).$$

The three main terms on the RHS are $O_P(1)$, $O_P(1/m)$, and $O_P(\sqrt{n}/m)$, so

$$T_{IV}(\hat{\tau}^2) = \sqrt{n/\hat{\tau}^2} P_n(Y) - \sqrt{n}/m(\hat{\tau}^2)^{-3/2} P_n(YV) + O_P(1/m) + O_P(\frac{\sqrt{n}}{m^2}).$$

Expanding the negative powers of $\hat{\tau}^2 = \tau^2 + P_n(Z) + O_P(1/m)$ about τ^2,

$$\begin{aligned}
T_{IV}(\hat{\tau}^2) &= n^{1/2} P_n(Y)(1/\tau - \frac{1}{2} P_n(Z)/\tau^3 + O_P(\frac{1}{m}) + O_P(\frac{1}{n})) \\
&\quad - n^{1/2} m^{-1} P_n(YV)(1/\tau^3 - \frac{3}{2\tau^5}(P_n(Z) + O_P(\frac{1}{m})) + O_P(\frac{1}{m}) + O_P(\frac{1}{n})) + O_P(\frac{1}{m}) + O_P(\frac{\sqrt{n}}{m^2}) \\
&= n^{1/2} P_n(Y)(1/\tau - \frac{1}{2} P_n(Z)/\tau^3 + O_P(\frac{1}{m})) - \frac{n^{1/2}}{\tau^3 m} P_n(YV) + O_P(\frac{1}{m}) + O_P(\frac{\sqrt{n}}{m^2}) + O_P(\frac{1}{\sqrt{n}}),
\end{aligned}$$

where the second equality uses the assumptions that $P_n(Z) = O_P(1/\sqrt{n})$ and $P_n(Y) = O_P(1)$. The first-order expansion follows by substituting $\frac{\sqrt{n}}{m} P_n(YV)P_n(Z) = O_P(\frac{1}{m})$, $\sqrt{n} P_n(Y)P_n(Z) = O_P(\frac{1}{\sqrt{n}}) + O_P(\frac{1}{m})$, and $\sqrt{n} P_n(Y) = \sqrt{n}(P_n(Y) - E_n(Y)) + \sqrt{n} E_n(Y)$. □

Proof of Proposition 3. The conclusions follow from the first-order expansion (16) for the IV statistic given in Lemma 7. □

Proof of Proposition 4. Since it has been assumed that $n = O(m)$, the second-order expansion (15) given in Lemma 7 has $O(1/n)$ error. It suffices to compute the Edgeworth expansion of this approximation (Bhattacharya et al., 1978; Skovgaard, 1981). As $m = m(n)$ changes with n, we use a triangular array Edgeworth expansion (Skovgaard, 1981), under which

$$\mathbb{P}((T_{IV}(\hat{\tau}^2) - \kappa_1)/\sqrt{\kappa_2} \leq x) = \Phi(x) - \frac{\kappa_3}{6\kappa_2^{3/2}}(x^2 - 1)\phi(x) + O_P(n^{-1}), \tag{17}$$

where κ_i denote cumulants of $T_{IV}(\hat{\tau}^2)$. The approximation to $T_{IV}(\hat{\tau}^2)$ may be written as

$$\begin{aligned}
T_{IV}(\hat{\tau}^2) &= A + \frac{\sqrt{n}}{m} B + O_P(m^{-1}) + O_P(n^{-1}) \qquad (18) \\
A &= \frac{\sqrt{n}}{\tau_0} P_n(Y) - \frac{\sqrt{n}}{2\tau_0^3} P_n(Y)P_n(Z) \\
B &= \frac{3}{2\tau_0^5} P_n(YV)P_n(Z) - \frac{1}{\tau_0^3} P_n(YV)
\end{aligned}$$

where under the given assumptions A and B are $O_P(1)$ and $\sqrt{n}/m$ is $o_P(1)$. Given a vector of random variables or vectors let $\overline{\kappa}$ denote the average cumulant, e.g., $\overline{\kappa}(Y) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{cum}(Y_i)$,

$\overline{\kappa}(Y,Z) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{cum}(Y_i, Z_i)$. . For univariate cumulants we sometimes use subscripts to indicate the order, e.g., $\kappa_1(Y_1)$ is the first cumulant of Y_1 and $\kappa_4(Z_1)$ is the fourth cumulant of Z_1. We make repeated use of linearity of the cumulant function in each of its arguments and the rule that the joint cumulant of r averages based on n independent random vectors is $O_P(n^{-(r-1)})$. For the third cumulant calculations below, we make repeated use of the Leonov-Shiryaev rules for decomposing generalized cumulants using integer partitions. See McCullagh (1987) for details. As in the previous proofs we assume without loss of generality that $\mu_0 = 0$.

We start with the first cumulant, i.e., the mathematical expectation of (18):

$$
\begin{aligned}
&\kappa(A + \frac{\sqrt{n}}{m}B) \\
&= \frac{\sqrt{n}}{\tau_0}\overline{\kappa}(Y) - \frac{\sqrt{n}}{2\tau_0^3}(\overline{\kappa}(Y,Z) + \overline{\kappa}(Y)\overline{\kappa}(Z)) + \frac{\sqrt{n}}{m}(\frac{3}{2\tau_0^5}(\overline{\kappa}(YV,Z) + \overline{\kappa}(YV)\overline{\kappa}(Z)) - \frac{1}{\tau_0^3}\overline{\kappa}(YV)) \\
&= \frac{\sqrt{n}}{\tau_0}\overline{\kappa}(Y) - \frac{\sqrt{n}}{2\tau_0^3}(\frac{1}{n}\overline{\kappa}(Y,Z) + \overline{\kappa}(Y)\overline{\kappa}(Z)) - \frac{\sqrt{n}}{m}\frac{1}{\tau_0^3}\overline{\kappa}(Y,V) \\
&= \frac{\sqrt{n}}{\tau_0}\overline{\kappa}(Y) - \frac{1}{\sqrt{n}}\frac{1}{2\tau_0^3}\overline{\kappa}(Y,Z) - \frac{\sqrt{n}}{\tau_0^3 m}\overline{\kappa}(Y,V).
\end{aligned}
$$

The second-to-last equality uses that $\overline{\kappa}(YV,Z)/n$ is $O_P(1/n)$, $\overline{\kappa}(YV)\overline{\kappa}(Z)$ is $O_P(1/m)$, and $\overline{\kappa}(Y)\overline{\kappa}(V)$ is $O_P(1/m)$. The last equality uses that $\overline{\kappa}(Y)\overline{\kappa}(V)$ is $O_P(1/m^2)$.

Next we compute the second cumulant, i.e., the variance, where linearity implies

$$\kappa_2(A + \frac{\sqrt{n}}{m}B) = \kappa(A,A) + 2\frac{\sqrt{n}}{m}\kappa(A,B) + \frac{n}{m^2}\kappa(B,B).$$

The first term on the RHS is

$$\kappa(A,A) = \kappa_2(\frac{\sqrt{n}}{\tau_0}P_n(Y)) - \frac{n}{\tau_0^4}\kappa(P_n(Y), P_n(Y)P_n(Z)) + \frac{n}{4\tau_0^6}\kappa_2(P_n(Y)P_n(Z)).$$

The first term on the RHS here is $\kappa_2(\frac{\sqrt{n}}{\tau_0}P_n(Y)) = \overline{\kappa}_2(Y)/\tau_0^2$. For the other two terms, since

$$
\begin{aligned}
\kappa(P_n(Y), P_n(Y)P_n(Z)) &= O_P(1/n^2) + \kappa_2(P_n(Y))\overline{\kappa}(P_n(Z)) + \kappa(P_n(Z), P_n(Y))\overline{\kappa}(P_n(Y)) \\
&= O_P(1/n^2) + \frac{1}{n}\overline{\kappa}_2(Y)\overline{\kappa}(Z) + \frac{1}{n}\overline{\kappa}(Y,Z)\overline{\kappa}(Y) \\
&= O_P(1/n^2) + O_P(\frac{1}{mn}) \\
\kappa_2(P_n(Y)P_n(Z)) &= \kappa(P_n(Y), P_n(Z), P_n(Y), P_n(Z)) + \kappa(P_n(Y), P_n(Y)P_n(Z))\kappa(P_n(Z)) \\
&\quad + \kappa(P_n(Z), P_n(Y)P_n(Z))\kappa(P_n(Y)) \\
&= O_P(1/n) + O_P(1/m),
\end{aligned}
$$

so that $\kappa(A,A) = \overline{\kappa}_2(Y)/\tau_0^2 + O_P(1/n) + O_P(1/m)$. In like manner, $\kappa(A,B) = O_P(1/\sqrt{n})$ and $\kappa(B,B) = O_P(1/n)$, giving the stated value for κ_2.

For the third cumulant, linearity implies

$$\kappa_3(A+\frac{\sqrt{n}}{m}B)=\kappa_3(A)+3\frac{\sqrt{n}}{m}\kappa(A,A,B)+3\frac{n}{m^2}\kappa(A,B,B)+\frac{n^{3/2}}{m^3}\kappa_3(B).$$

The cumulant $\kappa_3(B,B,B)$ consists of terms of the form $\kappa_3(P_n(YB)P_n(Z))$, $\kappa(P_n(YV)P_n(Z), P_n(YV)P_n(Z), P_n(YV))$, $\kappa(P_n(YV)P_n(Z), P_n(YV), P_n(YV))$, and $\kappa_3(P_n(YV))$. These may be simplified using the Leonov-Shiryaev rules as tabulated by the complementary partitions tables McCullagh (1987). In the partition notation there, we find for $\kappa_3(P_n(YB)P_n(Z))$ complementary partitions of the form 12|34|56, 12|34|5, 12|3|4, and 1|2|3, which all resolve to order $1/n^2$ for the RVs under consideration. Similarly, $\kappa(A,B,B)$ and $\kappa(A,A,B)$ are respectively $O(n^{-3/2})$ and $O(n^{-1})$. Therefore the third cumulant of $T_{IV}(\hat{\tau}^2)$ is the third cumulant of A,

$$\kappa_3=\kappa_3(A)+O_P(1/m)+O_P(1/n).$$

The latter is

$$\begin{aligned}\kappa_3(A)=&\frac{n^{3/2}}{\tau_0^3}\frac{1}{n^2}\overline{\kappa}_3(Y)-\frac{3n^{3/2}}{2\tau_0^5}\kappa(P_n(Y),P_n(Y),P_n(YZ))\\&+\frac{3n^{3/2}}{4\tau_0^7}\kappa(P_n(Y),P_n(YZ),P_n(YZ))-\frac{n^{3/2}}{8\tau_0^9}\kappa_3(P_n(Y)P_n(Z)).\end{aligned}$$

We show that all terms other than the first on the RHS are within the allowed error. For the second term on the RHS we look in the previously cited ables for partitions of the form 12|3|4 and find it resolves to

$$\begin{aligned}\kappa(P_n(Y),P_n(Y),P_n(YZ))&=\kappa_3(P_n(Y))\kappa(P_n(Z))+\kappa(P_n(Y),P_n(Y),P_n(Z))\kappa(P_n(Y))\\&\quad+2\kappa_2(P_n(Y))\kappa(P_n(Y),P_n(Z))\\&=\frac{1}{n^2}\overline{\kappa}(Y)\overline{\kappa}(Z)+\frac{1}{n^2}\overline{\kappa}(Y,Y,Z)\overline{\kappa}(Y)+\frac{2}{n^2}\overline{\kappa}_2(Y)\overline{\kappa}(Z,Y)\\&=O(n^{-5/2}).\end{aligned}$$

For the third term on the RHS, $\kappa(P_n(Y),P_n(YZ),P_n(YZ))$, we look up the partition form 12|34|5, finding it resolves to an $O_P(1/n^3)$ term; 4 partitions of the form 135|2|4 that are $O_P(1/n^2)O_P(1/m^2)$ and so also $O_P(1/n^3)$; and 8 partitions of the form 13|25|4 which are $O_P(1/n^2)O_P(1/m)=O_P(n^{-5/2})$. Lastly, for the fourth cumulant on the RHS, $\kappa_3(P_n(Y)P_n(Z))$, the tables give 13|25|4|6 as the complementary partition with the slowest rate, and the latter is $O_P(1/n^2)O_P(1/m^2)=O_P(1/n^3)$.

The values of κ_1 and κ_2 calculated above tend to 0 and 1 respectively as $n\to\infty$, and in fact under the null model assumptions and the assumption $n=O(m)$, $\kappa_1^2=O(1/n)$ while $\kappa_2-1\propto Var_n(Y)-\tau_0^2$ is $O(1/m)$. Expanding (17) with respect to (κ_1,κ_2) about $(0,1)$ then leads to

$$
\begin{aligned}
&\mathbb{P}(T_{IV}(\hat{\tau}^2) \leq x) \\
&\quad = \Phi(x) + \frac{\kappa_3}{6\kappa_2^{3/2}}(1-x^2)\phi(x) - (\kappa_1 + \frac{x}{2}(\kappa_2 - 1))(1 + \frac{\kappa_3}{6\kappa_2^{3/2}}(x^3 - 3x))\phi(x) + O_P(\frac{1}{m}) + O_P(\frac{1}{n}) \\
&\quad = \Phi(x) + \frac{\kappa_3}{6\kappa_2^{3/2}}(1-x^2)\phi(x) - \kappa_1(1 + \frac{\kappa_3}{6\kappa_2^{3/2}}(x^3 - 3x))\phi(x) + O_P(\frac{1}{m}) + O_P(\frac{1}{n}).
\end{aligned}
$$

□

Proof of Corollary 5. By Proposition 4,

$$
\begin{aligned}
&\mathbb{P}(|T_{IV}(\hat{\tau}^2)| \leq z_{1-\alpha/2}) = \mathbb{P}(T_{IV}(\hat{\tau}^2) \leq z_{1-\alpha/2}) - \mathbb{P}(T_{IV}(\hat{\tau}^2) < -z_{1-\alpha/2}) \\
&\quad = 1 - \alpha - \frac{\kappa_1\kappa_3}{3\kappa_2^{3/2}}(z_{1-\alpha/2}^3 - 3z_{1-\alpha/2})\phi(z_{1-\alpha/2}) + O_P(\frac{1}{m}) + O_P(\frac{1}{n}),
\end{aligned}
$$

and under the null model and moment bound assumptions of Prop. 4, $\kappa_1\kappa_3 = O_P(1/m) + O_P(1/n)$. □